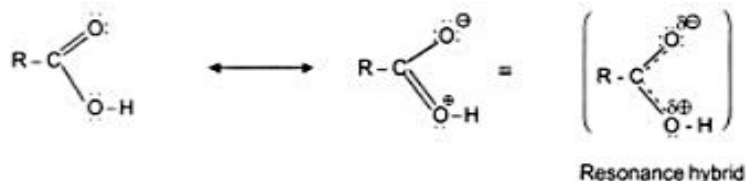


CARBOXYLIC ACID & ITS DERIVATIVES

GENERAL INTRODUCTION

The organic compounds containing carboxyl ($\text{-}\overset{\text{O}}{\parallel}\text{C}\text{-OH}$) group are known as carboxylic acids and the term 'Carboxylic' was firstly proposed by scientist 'Bayer'.

Resonance hybrid structure of carboxylic acid group is



Classification

On the basis of the group to which -COOH group is attached.

(a) Aliphatic carboxylic acid = $\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{OH}$ (R = H atom or alkyl group).

(b) Aromatic carboxylic acid = $\text{Ar}-\overset{\text{O}}{\parallel}\text{C}-\text{OH}$ (Ar = Aryl group)

On the basis of number of -COOH groups in their molecule.

	No. of -COOH group
(a) Mono carboxylic acids	1
(b) Dicarboxylic acids	2
(c) Tricarboxylic acids	3

General formula is $\text{C}_n\text{H}_{2n}\text{O}_2$ or $\text{C}_n\text{H}_{2n+1}\text{COOH}$

Saturated and unsaturated mono carboxylic acids are also called fatty acids, because higher acids like palmitic acid, stearic acid, oleic acid, etc. were obtained by hydrolysis of fats and oils.

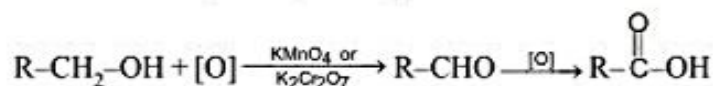
Isomerism – Chain, positional, functional group and optical.

Hybridisation state of C in the carboxyl group is sp^2

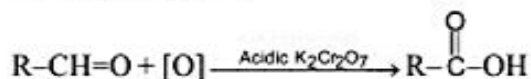
METHOD OF PREPARATION

By the oxidation of alcohols, aldehydes and ketones :

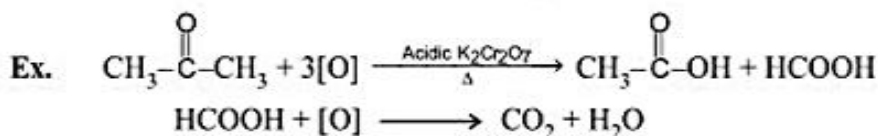
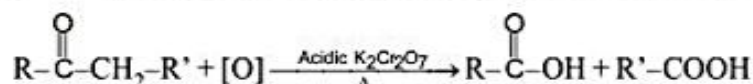
(a) 1° Alcohol $\xrightarrow{[\text{O}]}$ corresponding carboxylic acids.



(b) Aldehyde $\xrightarrow{[O]}$ corresponding acids

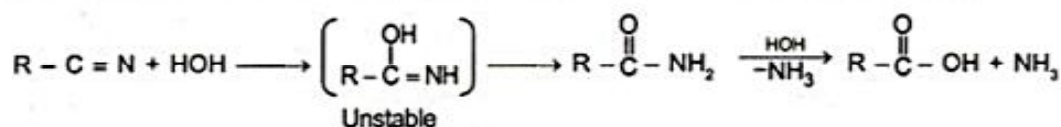


(c) Ketones $\xrightarrow{[O]}$ mixture of acids of lesser no. of C atoms than the ketone.



By the hydrolysis of cyanides (nitriles) :

Cyanides on complete hydrolysis in the presence of dilute HCl form carboxylic acids.



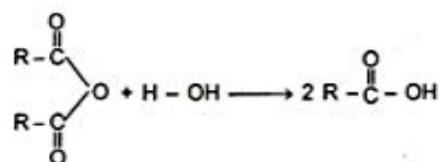
From hydrolysis of acid derivatives :



(a) **From acyl halides :**



(b) **From acid anhydride :**



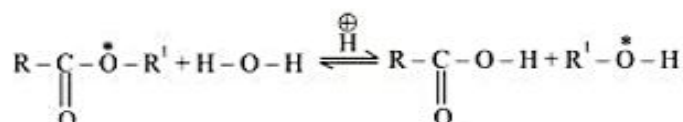
Note : Formic acid cannot be prepared by these two reactions (a & b) because its corresponding formyl chloride and formic anhydride are unstable compounds.

(c) **From carboxylic ester (Hydrolysis of Ester) :**

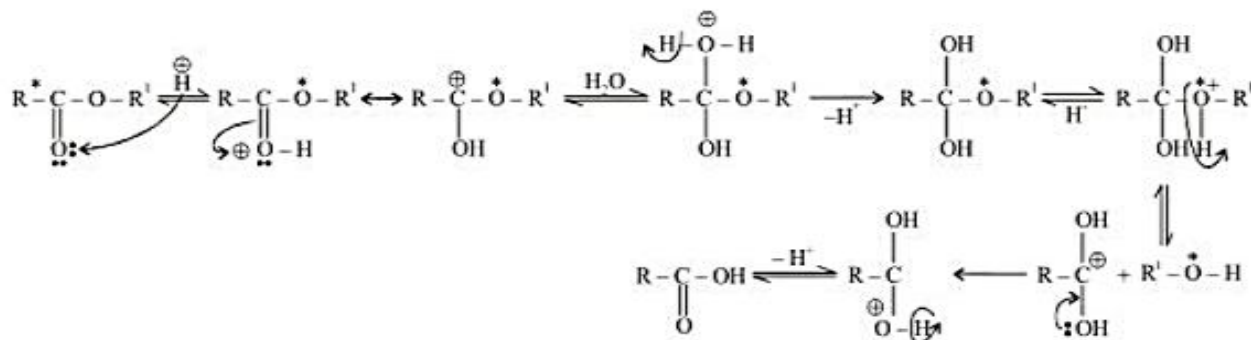


Hydrolysis of ester possible in acidic medium as well as in basic medium.

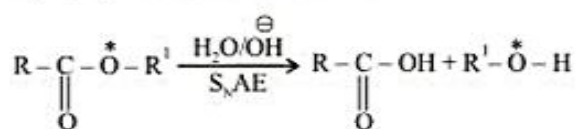
(i) **Hydrolysis in acidic medium :**



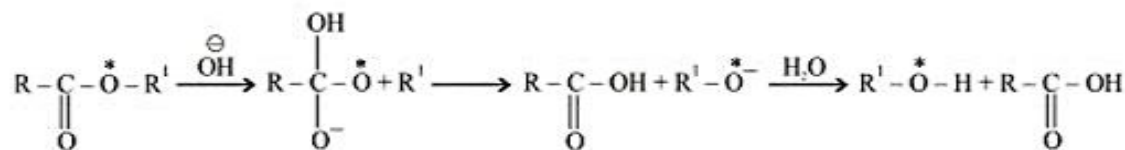
Mechanism :



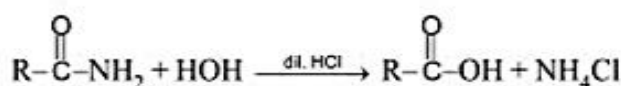
(ii) Hydrolysis in basic medium :



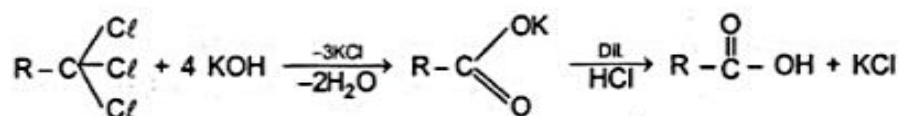
Mechanisms :



(d) From acid amides :

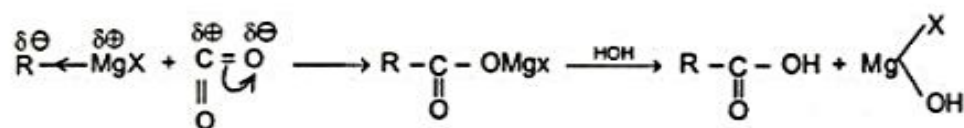


By alkaline hydrolysis of alkylidene trihalides :



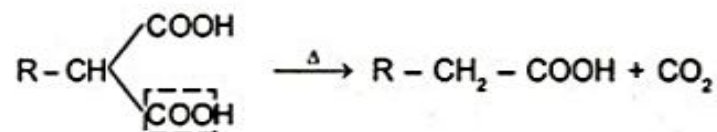
From carboxylation of grignard's reagent :

By passing CO_2 in ether Solution of Grignard's reagent

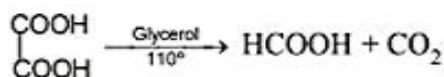


Note : Formic acid can not be prepared by this method.

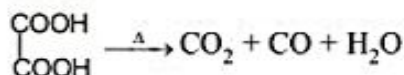
Partial decarboxylation of gem dicarboxylic acids :



Note : Formic acid is prepared from oxalic acid (laboratory method) by heating at about 110° in the presence of glycerol.

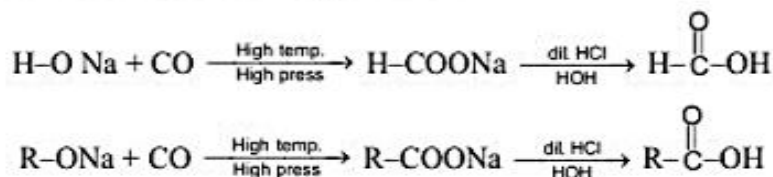


In absence of glycerol



Carboxylation of :

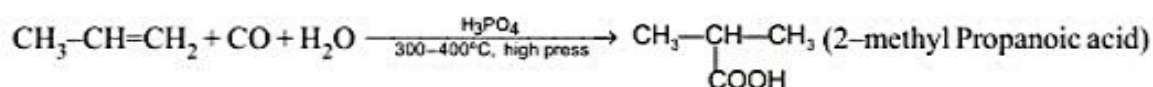
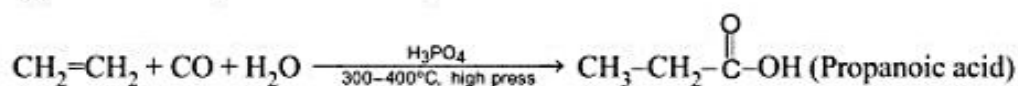
(a) **Sodium alkoxides or sodium hydroxide :**



(b) **Alcohols :**

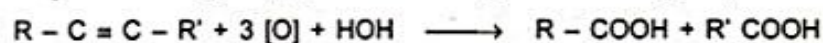
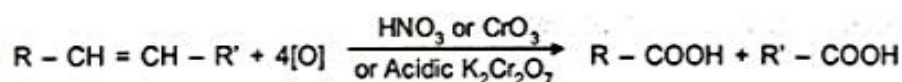


(c) **Alkenes [Koch Reaction] :**



Note : Formic acid and acetic acid can not be prepared by this method.

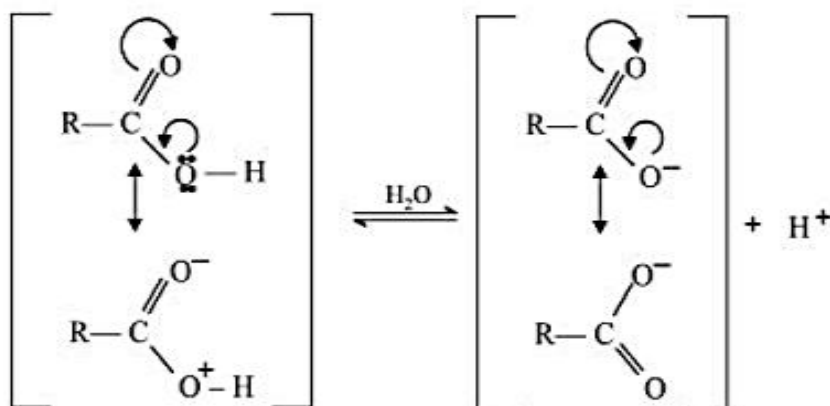
Oxidation of alkenes and alkynes :



CHEMICAL PROPERTIES

Acidic Nature

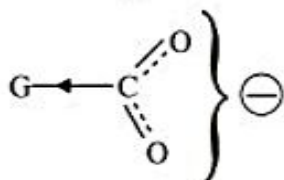
Carboxylic acids are weakly acidic (K_a values are of the order 10^{-4} to 10^{-5}) but a measurable equilibrium exists between acids and ions.



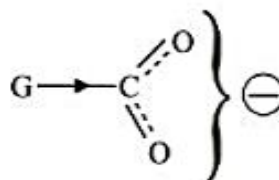
Both the acid and anion are resonance stabilised, but stabilisation is more powerful between exactly equivalent resonating structures of the carboxylate ion.

Effect of Substituents on Acidity

Electron-withdrawing groups disperse the negative charge on the carboxylate ion, thereby stabilising it and increasing the acidity. Electron-donating groups intensify the negative charge, thereby destabilising the carboxylate ion and decreasing the acidity.



G withdraws electrons: stabilises anion, strengthens acid



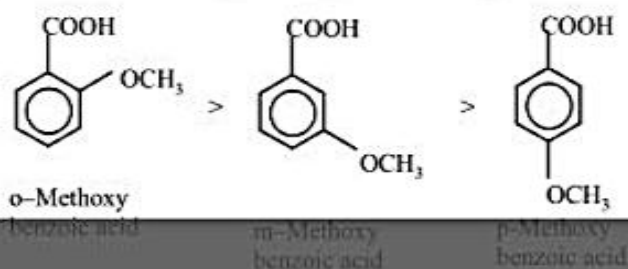
G releases electrons: destabilises anions, weakens acid.

For example, chloroacetic acid is hundred times stronger than acetic acid.

The effect of substituents on acidic strength of benzoic acids is the same. For example, benzoic acid is stronger than p-methoxybenzoic acid but weaker than p-nitrobenzoic acid. (Methoxy group is electron-releasing whereas nitro group is electron-withdrawing.)

Ortho Effect

Among isomeric substituted benzoic acids, ortho-isomer is the strongest acid irrespective of the nature of substituent. Example: The acidic strength among the following acids is of the order

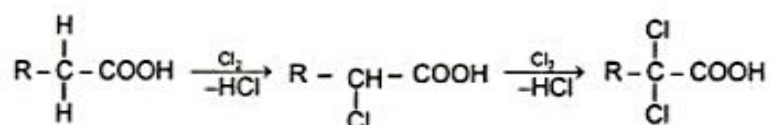


Carboxylic acids give following reactions :

1. Due to alkyl group
2. Due to acidic hydrogen
3. Due to $>C=O$ group
4. Due to $-OH$ group
5. Due to $-COOH$ group
6. Special reaction of alkanolic acid

(1) Reactions due to alkyl group :

Halogenation : Acids having $\alpha-H$ atom give this reaction. The $\alpha-H$ atoms are easily replaced by chlorine or bromine atoms, forming halogen derivatives of acids.

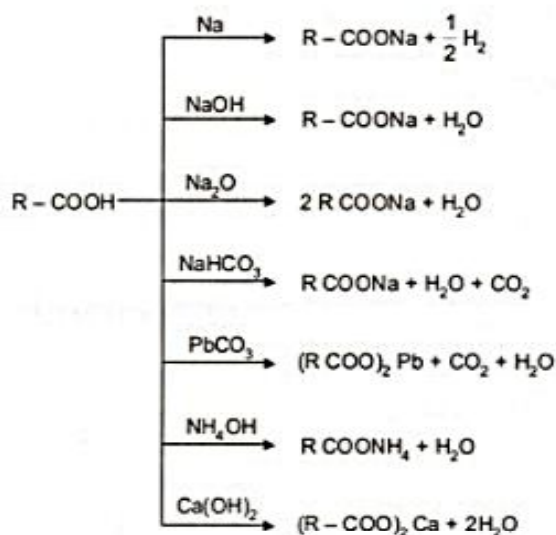


- Note :**
- (i) The reaction is known as hell-volhard-Zelinisky (HVZ) reaction.
 - (ii) In the presence of phosphorus, monohalo product is formed as the main product.

$$\text{R}-\text{CH}_2-\text{COOH} + \text{Br}_2 \xrightarrow{\text{P}} \text{R}-\text{CH}(\text{Br})-\text{COOH} + \text{HBr}$$
 - (iii) Formic acid does not show HVZ reaction due to absence of alkyl group or $\alpha-H$ atom.

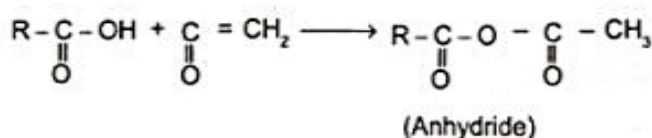
(2) Due to acidic hydrogen :

- (a) Salt Formation :** Carboxylic acids form salts with metals, metal oxide, hydroxides, carbonates, bicarbonates and compounds showing basic nature.

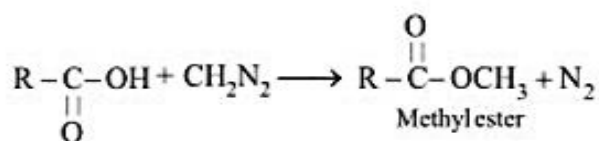


Note : Acids liberate CO_2 with NaHCO_3 . This reaction is used to distinguish carboxylic acid from phenol which does not react with NaHCO_3 .

(b) Reaction with Ketene :

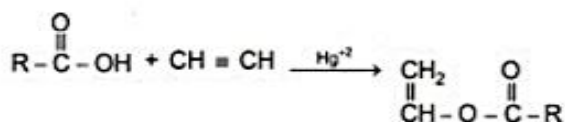
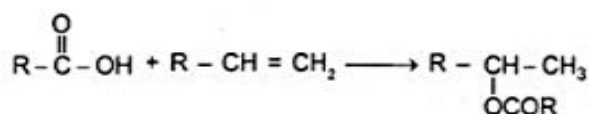
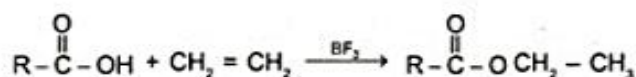


(c) **Reaction with diazomethane :**

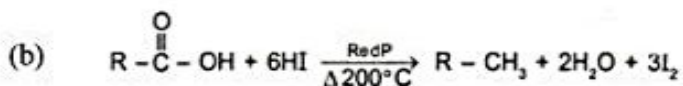
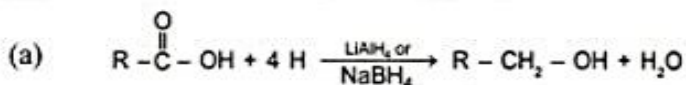


Note : Diazomethane is a powerful methylating agent. So by this method methyl ester can be prepared.

(d) **Reaction with alkene and alkyne :**

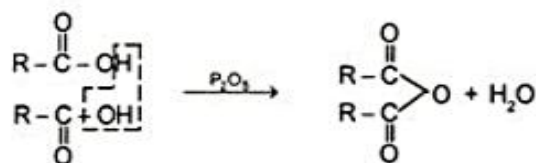


(3) **Reaction due to (>C=O) group :**

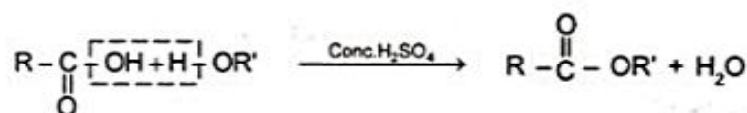


(4) **Reactions due to (-OH) group :**

(a) **With P₂O₅ (Dehydrations)**

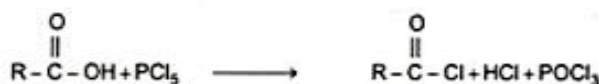


(b) **Esterification :**

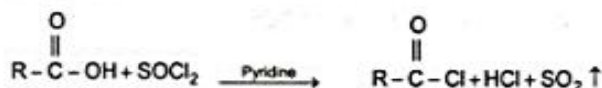


Note : The reaction is proton-catalysed esterification. In the reaction OH comes from the acid and H from the alcohol for release of H₂O.

(c) **Reaction with Phosphorus Chlorides :**

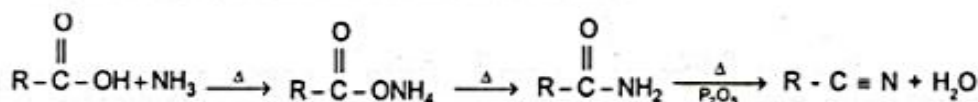


(d) **With Thionyl Chloride :**

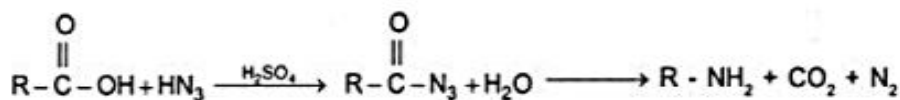


Note : In the reaction SO_2 is gas so escapes out and HCl is an acid, gets absorbed by basic pyridine and pure acyl chloride is left. So its is best method for preparing acyl chloride from acids.

(e) **Formation of acid Amides and Alkane Nitriles :**



(f) **With Hydrazoic acid (Schmidt reaction) : N_3H = Hydrazoic acid**

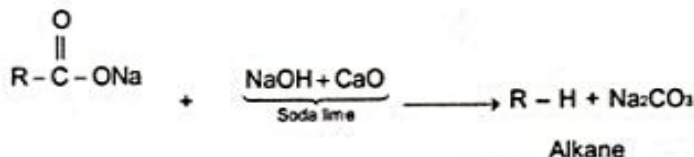


The reaction is a modification of "curtius reaction".

(5) **Reaction due to $(-\text{COOH})$ group :**

(a) **Decarboxylation of Carboxylic Acids**

On heating carboxylic acids with soda-lime (NaOH and CaO), an alkane is obtained with one carbon less than the carboxylic acids.



Mechanisms :

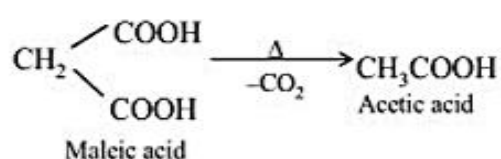
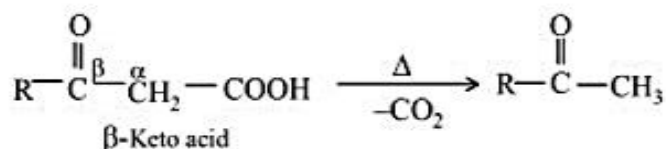


Note :-

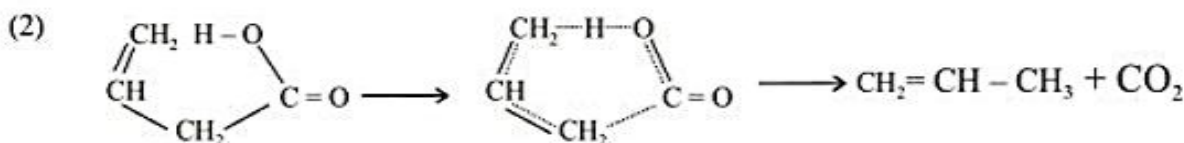
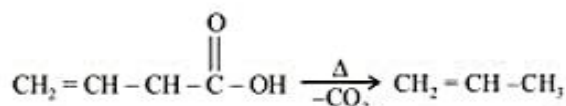
- (i) Step Ist is reversible but presence of CaO shift the equilibrium in the formed direction.

$$\text{CaO} + \text{CO}_2 \longrightarrow \text{CaCO}_3$$
- (ii) Step Ist is RdS i.e. formation of carbanion
- (iii) Rate of decarboxylation is directly proportional to stability of carbanion.

β -keto acid : β -Keto acids decarboxylate readily when heated to a temperature of $100^\circ\text{C} - 150^\circ\text{C}$.

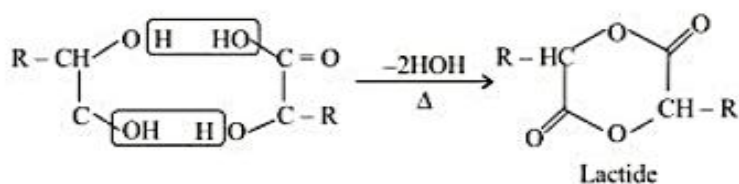


- (1) β - γ unsaturated carboxylic acid gives alkene on simple heating.

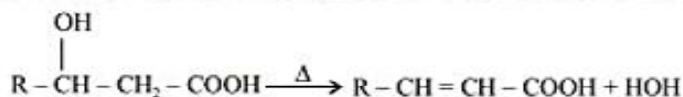


Heating effect

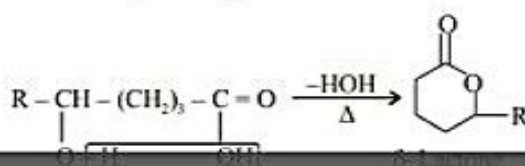
α -Hydroxy acid : α -hydroxy acids on heating form lactide. Reaction takes place between two moles of acid.



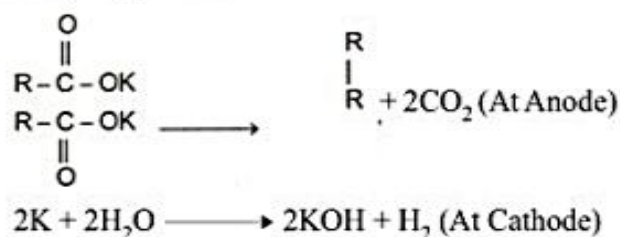
β -Hydroxy acid : β -hydroxy acids on heating undergo dehydration to give α, β unsaturated acids.



γ -and higher hydroxy acids : They form cyclic ester known as lactones.



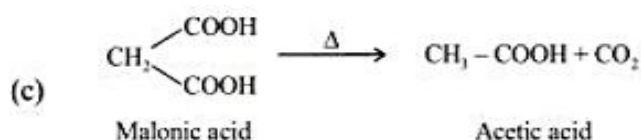
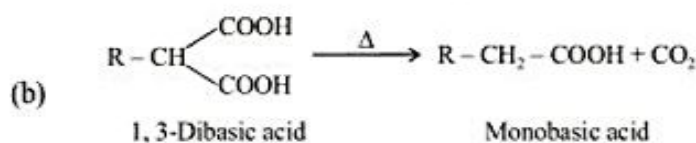
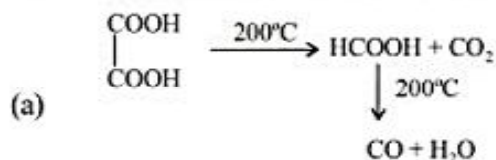
(b) **Kolbe's electrolytic synthesis :**



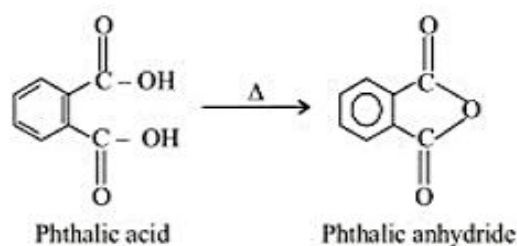
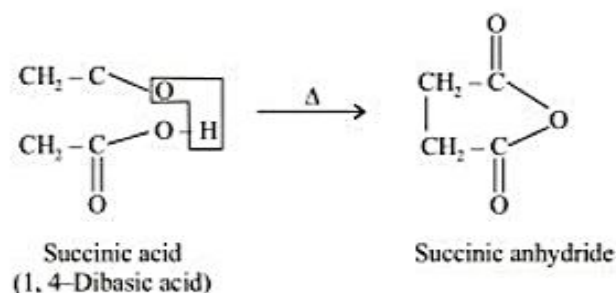
Action of heat or dehydrating agents on dibasic acids

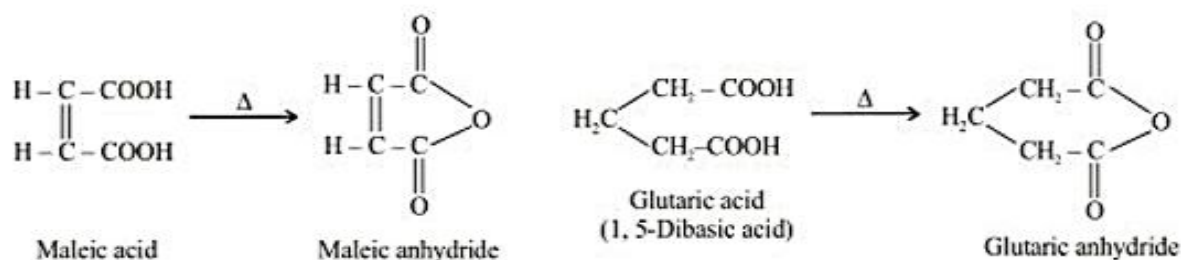
Variety of products are obtained when dicarboxylic acids are heated. The nature of product depends on the carbon chain length separating the two carboxylic groups.

(1) **1, 2 and 1, 3-dibasic acids undergo decarboxylation to form monobasic acids.**

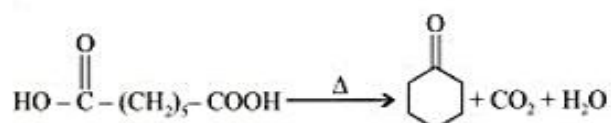


(2) **1, 4 and 1, 5-dibasic acid undergo intramolecular dehydration to give cyclic anhydrides.**



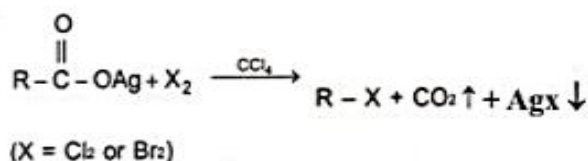


- (3) **1, 6 and 1-7dibasic acid :** These two acids undergo dehydration as well as decarboxylation to give cyclic ketones. 1, 6-Dibasic acids give cyclopentanone and 1, 7-dibasic acids give cyclohexanone.



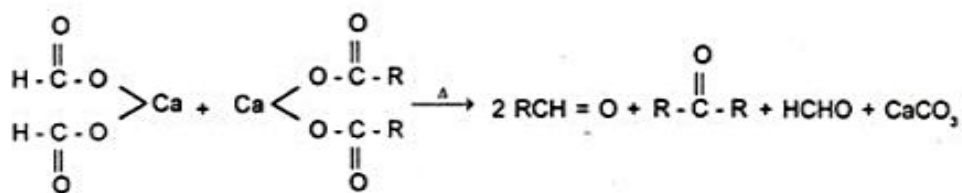
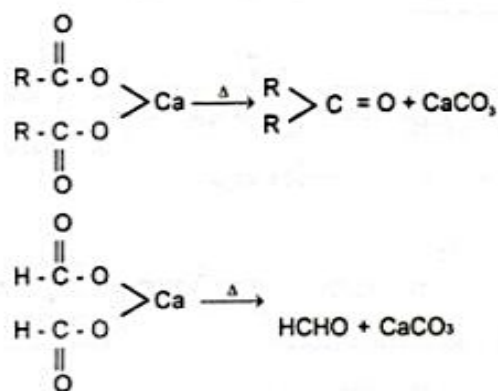
Note : Kolbe synthesis undergoes free radical mechanism.

(c) **Hunsdiecker reaction :**

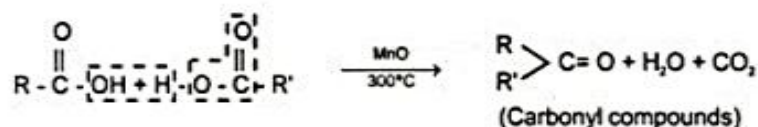


Note : In the reaction decarboxylative halogenation of the acid takes place. The reaction is useful in stepping down a homologous series.

(d) **Formation of carbonyl compounds [Dry distillation of calcium salt] :**



(e) **With Manganous oxide :**

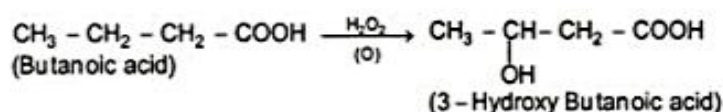


(5) **Special reaction of alkanolic acid :**

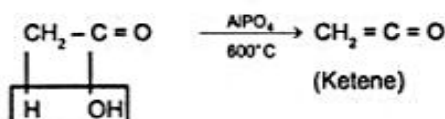
(a) **Oxidation :**

→ By SeO_2 α keto acid is formed

→ By H_2O_2 -OH group comes at β -C atom.



(b) **Pyrolysis :**



SPECIAL FEATURES OF FORMIC ACID

(a) Formic acid shows acidic character due to the presence of -COOH group as well as shown reducing character due to the presence of -CHO group.



(b) It reduces Tollen's reagent and forms silver mirror.

(c) It reduces mercuric chloride and forms black ppt. of mercury.

(d) It decolourises the pink colour of acidic KMnO_4

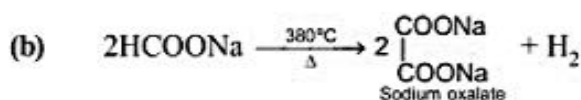
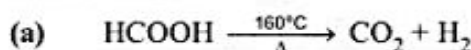
(e) It forms brown ppt. of MnO_2 with basic KMnO_4

(f) It converts the orange colour of acidic $\text{K}_2\text{Cr}_2\text{O}_7$ into green colour.

(g) It reduces Fehling solution.

(h) Dehydration $\text{HCOOH} \xrightarrow[\text{Conc. H}_2\text{SO}_4]{\Delta} \text{H}_2\text{O} + \text{CO}_2$

(i) **Effect of heat :**

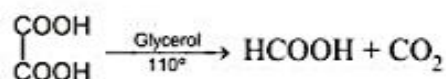


DISTINCTION BETWEEN FORMIC ACID AND ACETIC ACID

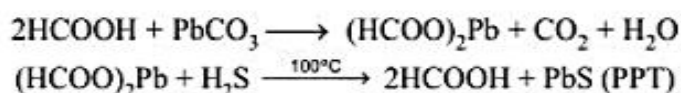
S.No.	HCOOH	CH ₃ COOH
1.	Decomposed on heating to give carbon dioxide and hydrogen	Stable
2.	With concentrated sulphuric acid gives carbon monoxide and water	Stable
3.	With halogen, there is not action	Substituted acids are formed.
4.	Reduces ammonical silver nitrate and Fehling's solution	No action
5.	Calcium salt on heating gives formaldehyde	Calcium salt on heating gives acetone
6.	Sodium salt on heating gives sodium oxalate hydrogen	No action
7.	Sodium salt + Soda lime gives hydrogen	Methane is produced
8.	Electrolysis of alkali solution gives hydrogen	Ethane is obtained
9.	Decolourises acidified KMnO ₄ solution	No action
10.	Dichromate solution turns into green coloured solution	No action

FORMIC ACID (HCOOH)

1. Lab Reaction :



Note : BP of HCOOH is 100.5°, for removal of water from formic acid we can use PbCO₃ or PbO (Litharge) and H₂S.



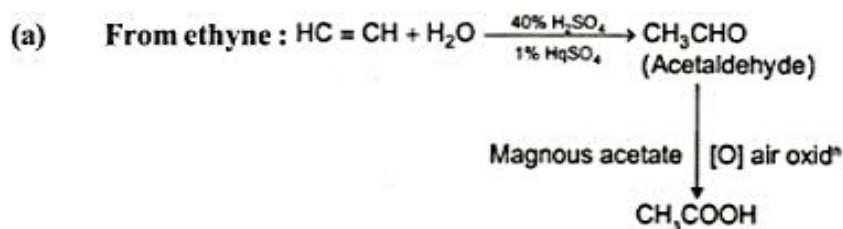
2. Industrial method :

It is prepared on large scale by reaction CO with aqueous sodium hydroxide under pressure 1 atm and 473 K

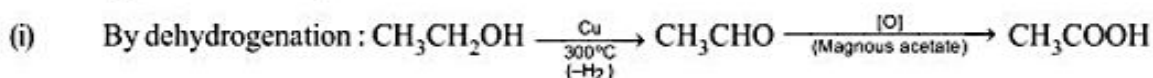


ACETIC ACID (CH_3COOH)

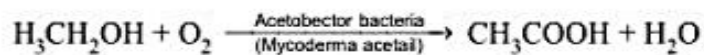
Industrial methods



(b) From ethyl alcohol

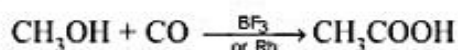


(ii) By fermentation (Quick vinegar process)



[dilute form 8 – 10% Called VINEGAR]

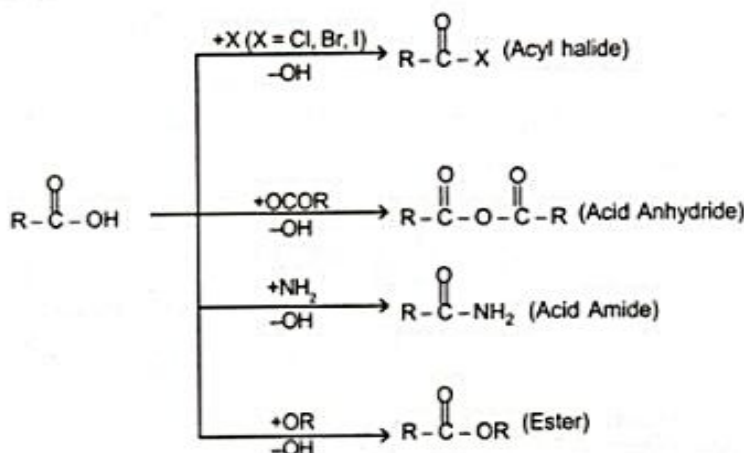
(iii) From CH_3OH



ACID DERIVATIVES

GENERAL INTRODUCTION

Replacement of $-OH$ group of the $-COOH$ group in a carboxylic acid by an atom or by any other functional group give rise to certain new families of compounds. These compounds are called acid derivatives.

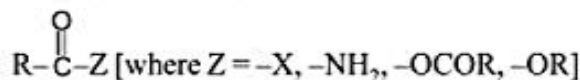


Acyl chlorides are the most important acid halides. The first member of this family is acetyl chloride because formylchloride is unstable.

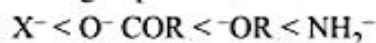
Similarly acetic anhydride is the representative member of the acid anhydrides family, because formic anhydride is unstable.

The first two members of Acid Amide family are formamide and Acetamide. Acetamide is an important acid amide.

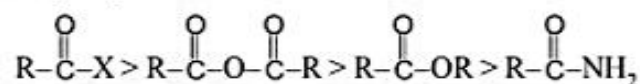
Esters have functional group isomerism with acids.



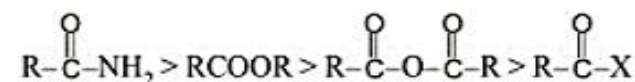
* Order of basic character of group Z



* Reactivity order/Rate of hydrolysis



* Order of stability

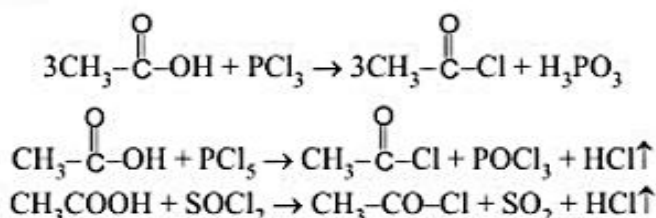


Derivatives regenerate the parent acid on hydrolysis.

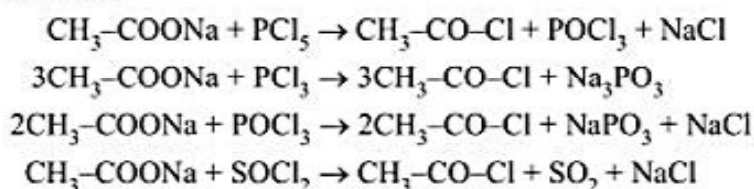
ACETYL CHLORIDE $[\text{CH}_3\text{-}\overset{\text{O}}{\parallel}\text{C}\text{-Cl}]$

Method of preparation :

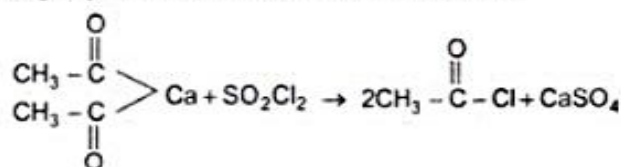
Laboratory method :



From Sodiumacetate :



By the reaction of sulphuryl chloride on calcium ethanoate :



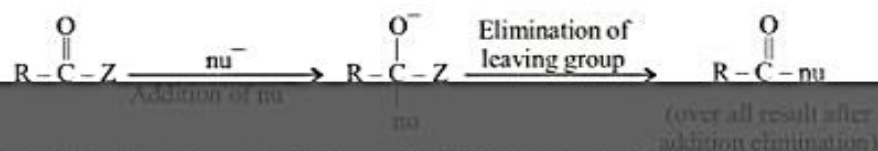
Physical Properties :

- Colourless liquid, having pungent odour.
- Slowly soluble in water, soluble in organic solvents like CHCl_3 , CCl_4 , $\text{CH}_3\text{-CO-CH}_3$, etc.
- Produces blister on skin and its vapour causes irritation in eyes.
- Boiling point is 51°C .
- Fumes strongly in moist air.

S_{NAE} Reaction : If attacking reagent is nucleophile then most of the reaction of acid derivative (besides $\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{NH}_2$) take place via S_{NAE} mechanism. S_{NAE} stands for nucleophilic addition elimination substitution reaction.



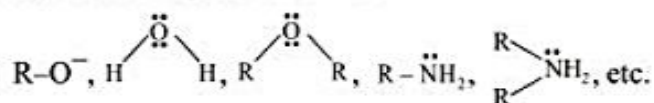
Mechanisms :



Saved /storage/emulated/0/Pictures/TouchShot/
20170809_000835.jpg

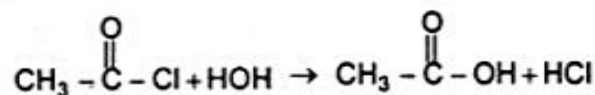
Here

Note : nu^- may be OH^- , NH_2^-

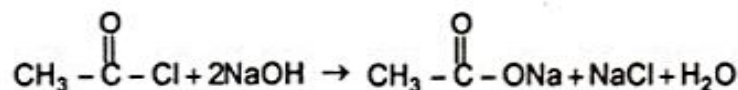


Chemical properties :

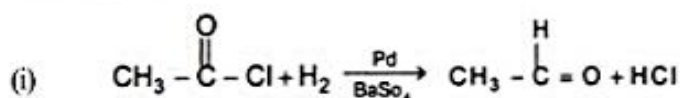
(1) **Hydrolysis :**



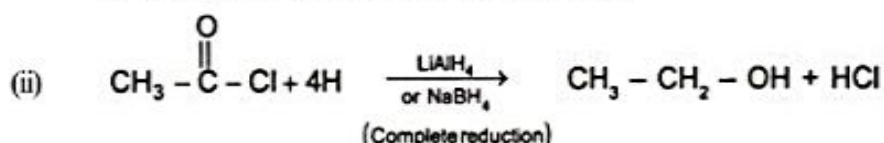
on alkaline hydrolysis



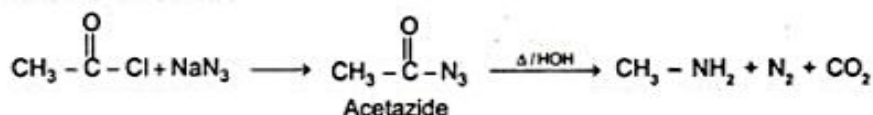
(2) **Reduction :**



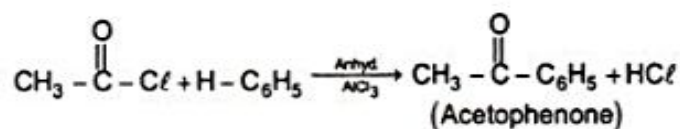
(Rosenmund's reduction, partial reduction)



(3) **Curtius Reaction :**



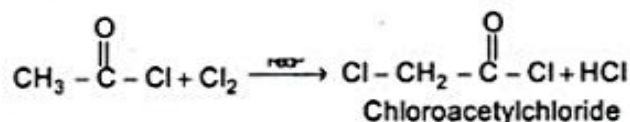
(4) **Friedel Craft's reaction :**



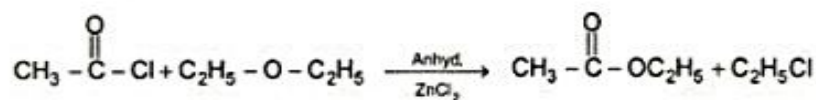
(5) **With Grignard's reagent :**



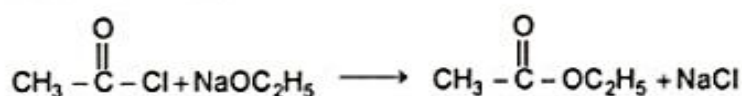
(6) **Halogenation :**



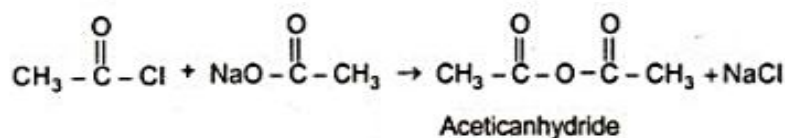
(7) **With Ether :**



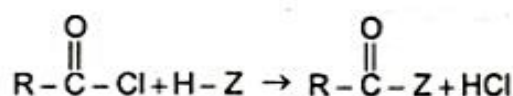
(8) **With Sodium Ethoxide :**



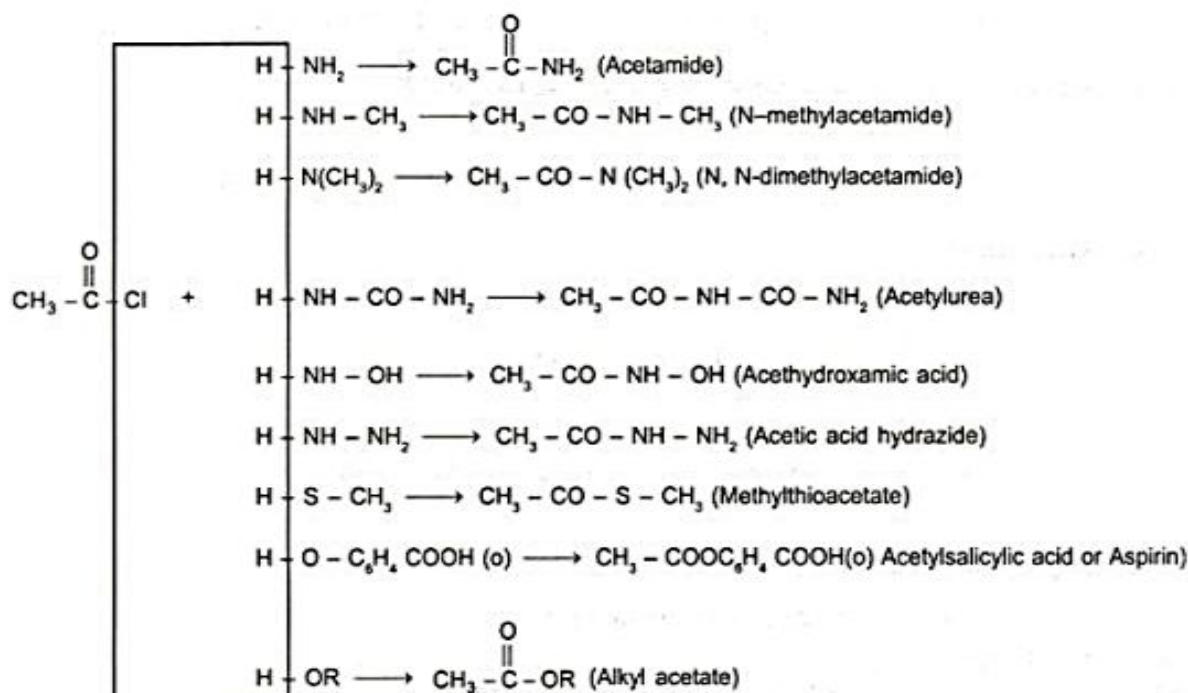
(9) **With Sodium Acetate :**



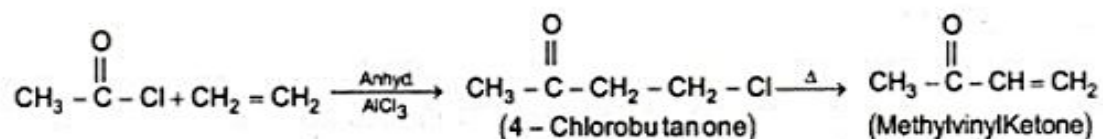
(10) **Acetylation of Compounds having reactive hydrogen atom :**



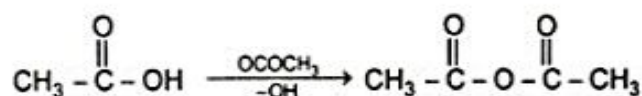
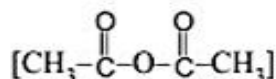
Ex.



(11) With Alkene :

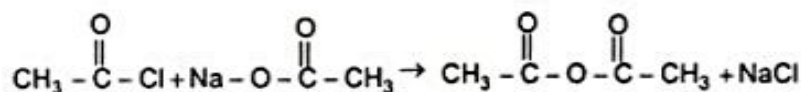
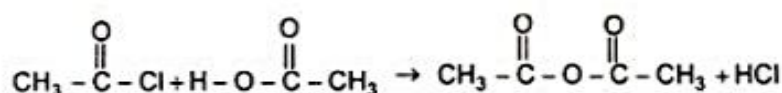


ACETIC ANHYDRIDE



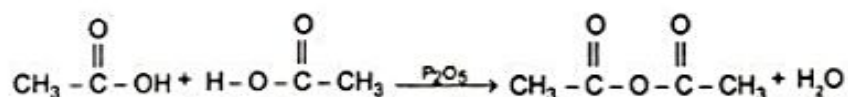
(1) Method of preparation :

From Acetyl chloride :

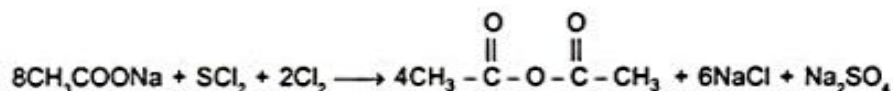


(Laboratory Method)

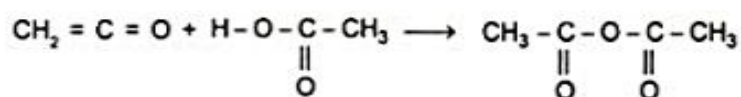
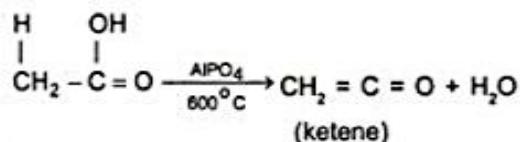
From Acetic acid :



By passing chlorine in a mixture of SCl_2 and sodium acetate :



From Ketene :

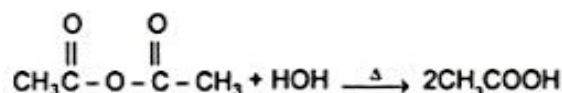


(2) Physical Properties :

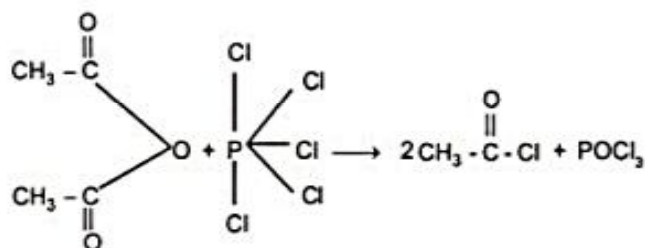
- (i) It is colourless liquid having pungent vinegar like odour.
- (ii) Boiling point is 138°C.
- (iii) Slowly soluble in cold water, but readily soluble in hot water

(3) Chemical Properties :

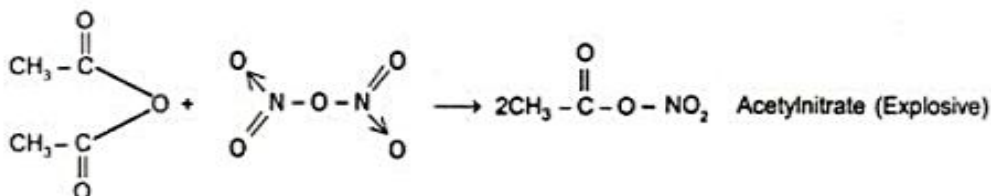
Hydrolysis :



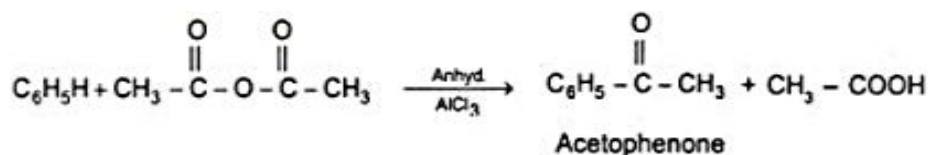
With PCl_5 :



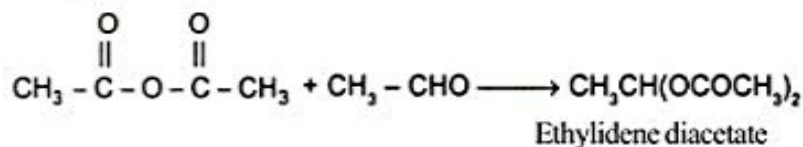
With N_2O_5 :



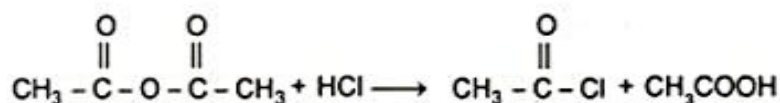
With Benzene [Friedel–Craft's Reaction] :



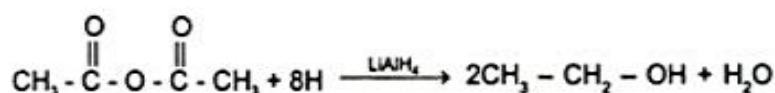
With Acetaldehyde :



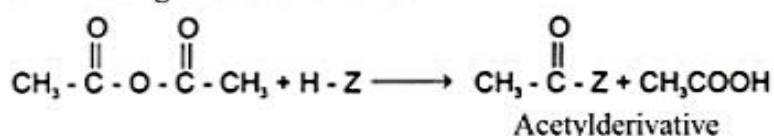
With Dry HCl gas :



Reduction :

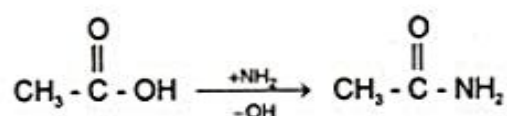
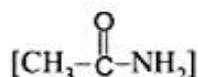


With Compounds having reactive H-atom :



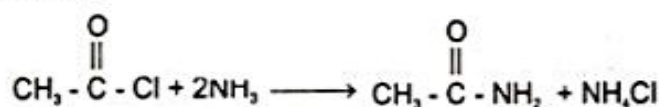
Note : Products are similar as in case of Acetyl chloride.

ACETAMIDE

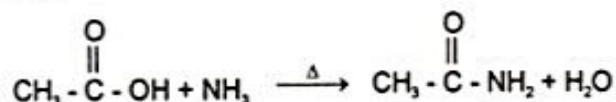


(1) Method of Preparation :

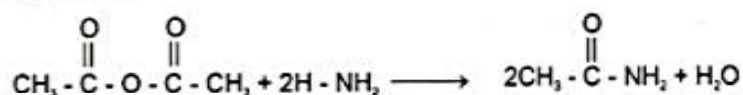
From Acetyl chloride :



From Acetic Acid :



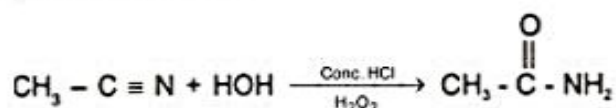
From acetic Anhydride :



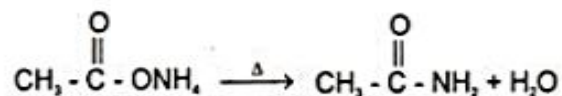
From Ethyl Acetate :



Partial hydrolysis of Acetonitrile :



By thermal decomposition of Ammonium acetate :

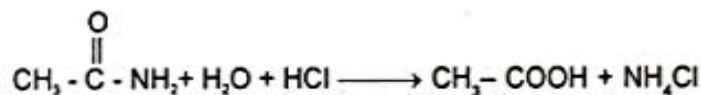


(2) **Physical properties :**

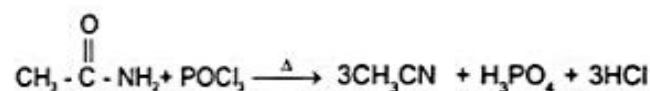
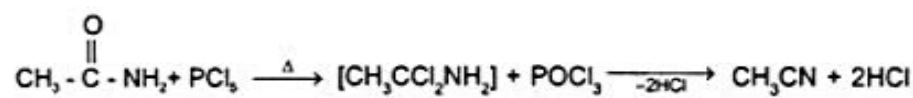
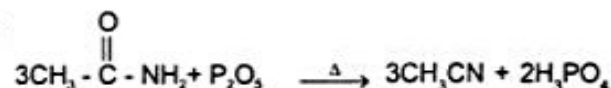
- (i) It is a hygroscopic white crystalline solid.
- (ii) It has faint smell in pure state, but a rat like smell in impure state.
- (iii) It is soluble in water and alcohol, but sparingly soluble in ether.
- (iv) Melting point 82°C and boiling point is 222°C.

(3) **Chemical properties :**

Hydrolysis :

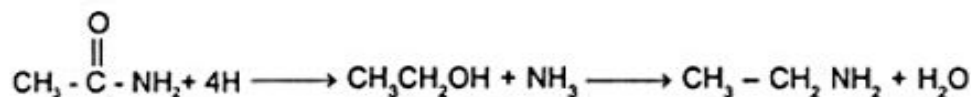


Dehydration :



Reduction :

Reducing Agent : Metal Catalyst + H₂, Na + Ethanol, LiAlH₄ etc.



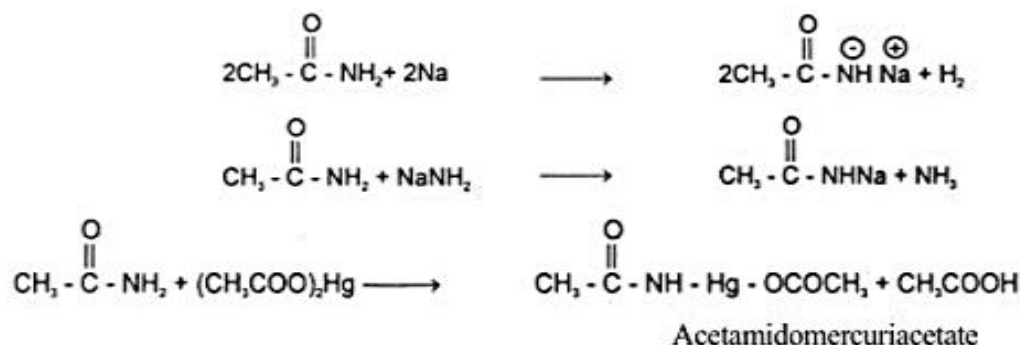
With Nitrous Acid :



Amphoteric character :

Acetamids forms salts on reacting with strong bases as well as strong acids. Therefore it behaves like a weak base as well as weak acid. i.e. it shows amphoteric character.

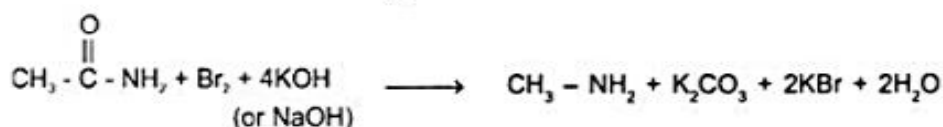
(i) **Weak acidic behaviour :**



(ii) **Weak basic behaviour :**

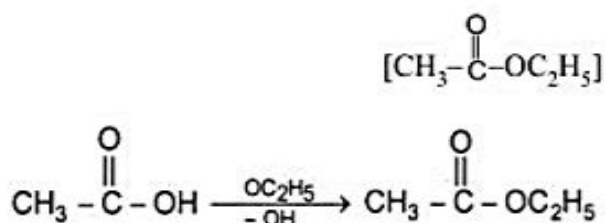


Hofmann Bromamide Reaction or Hypobromite Reaction :



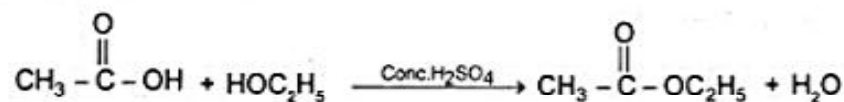
Note : In the reaction decrease of one carbon atom takes place. Therefore it can be used in descending down a homologous series and it is also called Hofmann degradation.

ETHYL ACETATE AND ACETIC ESTER

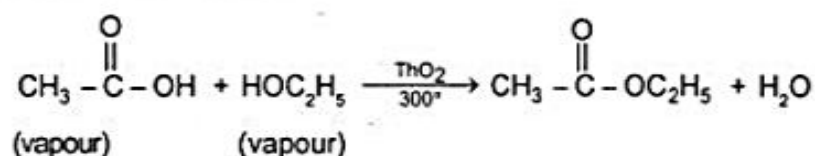


(1) **Methods of preparation :**

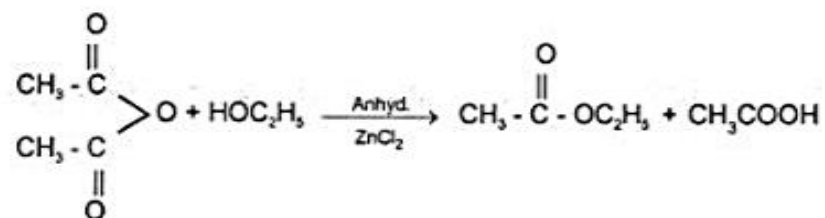
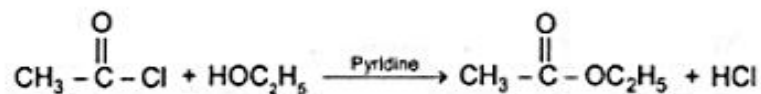
By liquid phase Esterification :



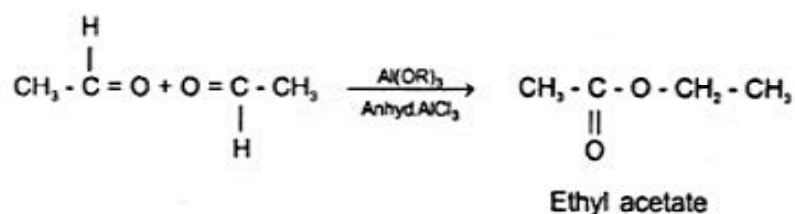
By vapour phase Esterification :



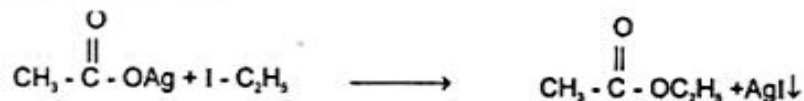
By Aacylation of Ethanol :



Tischenko Reaction :



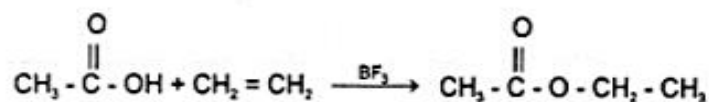
Ethylation of Silver halides :



Ethanolysis of Ethanenitrile :



By addition of Acetic acid on Ethylene :

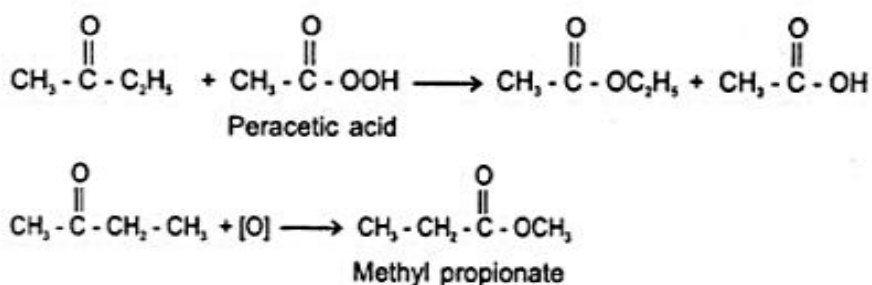


By reaction of sodium Ethoxide on Acetyl chloride :



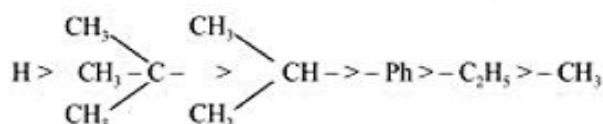
Baeyer-Villiger Oxidation :

Ethyl acetate is formed on reacting butanone with a peracid, Methyl propionate is also formed in small amounts.



Note : (Direct approach)

- (I) In Baeyer villiger oxidation 'O' is inserted between $-\overset{\text{O}}{\parallel}\text{C}$ & alkyl group.
- (II) If alkyl groups are different then 'O' is inserted between $-\overset{\text{O}}{\parallel}\text{C}$ & that alkyl group which has greater migratory aptitude.
- (III) Order of migratory aptitude in Baeyer villiger Oxidation is

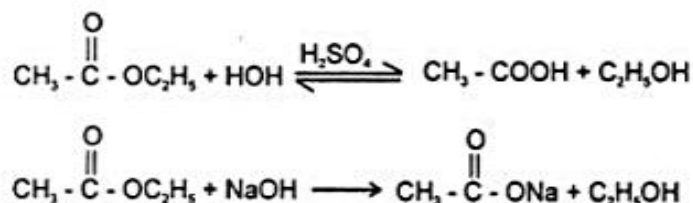


(2) Physical Properties :

- (i) Colourless, sweet smelling steamvolatile, inflammable liquid.
- (ii) Very sparingly soluble in water, but soluble in organic solvents.
- (iii) Boiling point 78°.

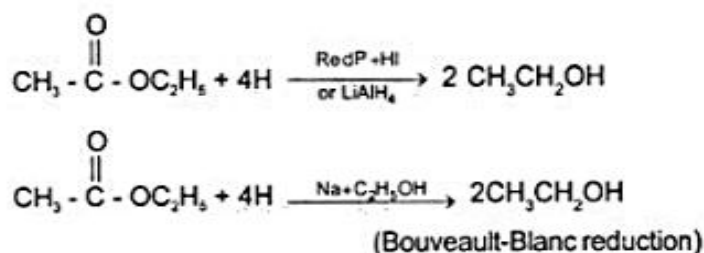
(3) Chemical Properties :

Hydrolysis :

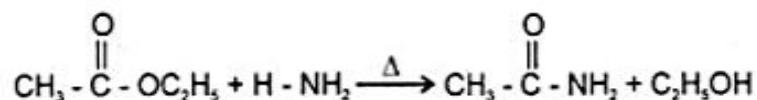


Note : In esters caustic alkalies hydrolysis is rapid and complete. Alkaline hydrolysis of esters is called "Saponification".

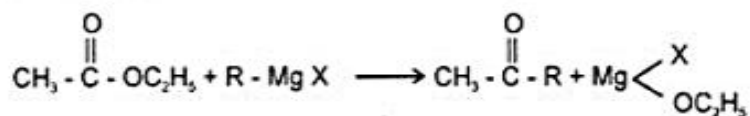
Reduction :



Ammonolysis :

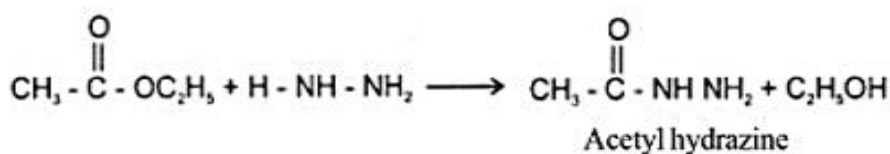


With Grignard's Reagent :

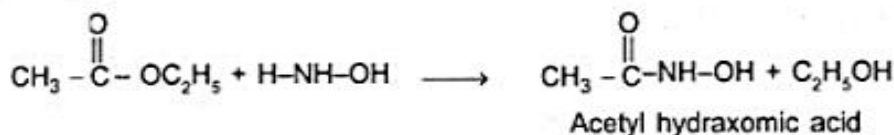


Note : The product will be tertiary alcohol when 2 mole of Grignard's reagent are taken.

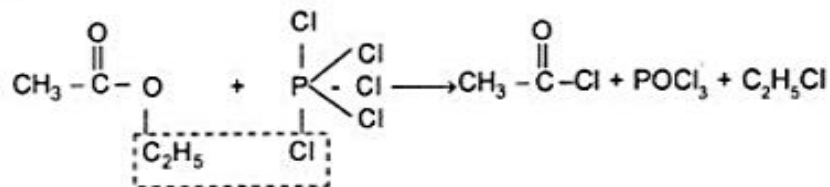
With Hydrazine :



With Hydroxyl Amine :

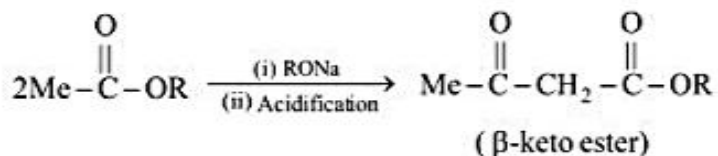


With PCl_5 :

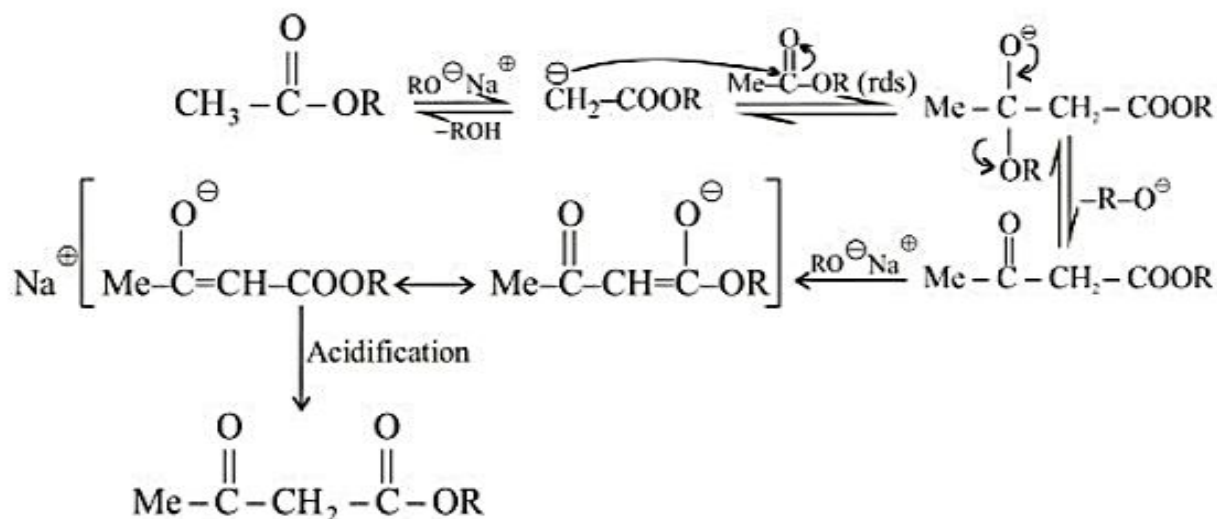


Claisen ester Condensation

Esters undergo $\text{S}_{\text{N}}\text{AE}$ Reaction, when attacked by a $\text{Nu}^\ominus$ generated by the interaction of a base (usually base related to the Alkoxy anion of ester) with one of the molecule of ester and this $\text{Nu}^\ominus$ attacks on another molecule. The reaction over all is considered as condensation of esters known as claisen ester condensation.

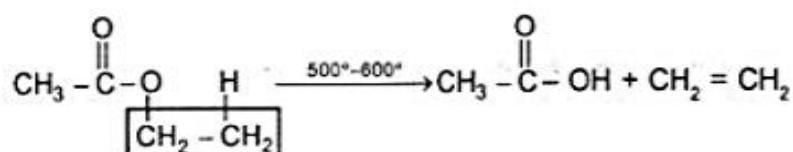


Mechanism :



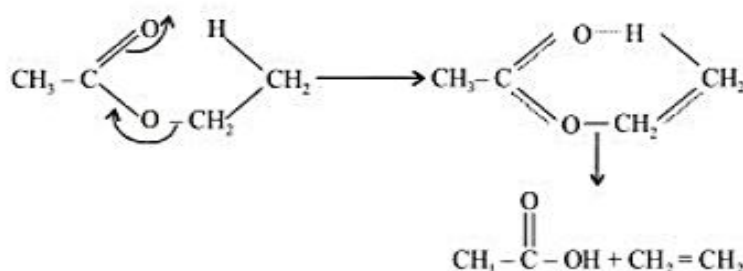
Some times, when two ester groups are present within the molecule then the condensation occurs intramolecular then cyclization caused thus is known as Dieckmann cyclization or Dieckmann's condensation.

Pyrolysis or Thermal decomposition :

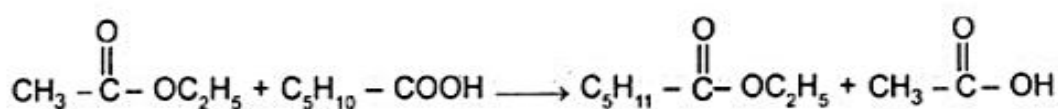


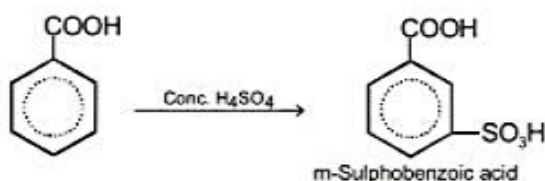
Mechanisms

Reaction takes place via 6 membered cyclic transition state



Acidolysis :





Note : Due to the presence of a deactivating m-directing COOH group in benzoic acid, the latter does not give Friedel–Crafts reaction.

(5) Tests

1. Formation of Ferric salt :

A red–brown precipitate is obtained on adding a small amount of neutral ferric chloride solution in benzoic acid solution.

2. Esterification :

A sweet odour of ethyl benzoate is perceived on heating benzoic acid after adding a few drops of ethyl alcohol and concentrated sulphuric acid.

3. Decarboxylation :

Inflammable vapours of benzene are liberated out on heating benzoic acid after mixing it thoroughly with soda lime.

(6) Uses

Benzoic acid is used as follows :

1. As a germicide.
2. Benzoic acid and its salts are used in the form of medicines.
3. Sodium benzoate is used as a preservative of canned food products.
4. Benzoic acid is used in the manufacture of many synthetic dyes, like aniline blue dye.
5. Ethyl ester of benzoic acid (ethyl benzoate) is used as a flavouring agent.

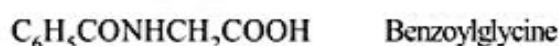
BENZOIC ACID

BENZOIC ACID (C_6H_5COOH)

Benzoic acid was first obtained by Schiele from gum benzoin. It is also present in Peru balsam and Tolu balsam in the form of its benzyl ester (benzyl benzoate).



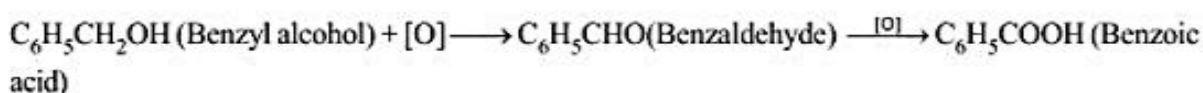
In combined form it is found as benzoylglycine in horse urine.



(1) Preparation

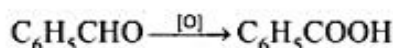
(1) Oxidation of Benzyl Alcohol :

Benzoic acid is formed on permanganate oxidation of benzyl alcohol.



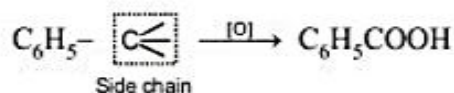
(2) Oxidation of Benzaldehyde :

Oxidation of benzaldehyde takes place very readily and benzoic acid is formed.



(3) Oxidation of Aralkanes and Their Derivatives :

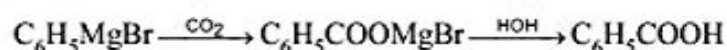
A side chain (alkyl group or substituent alkyl group) bonded directly to the benzene ring is converted to a carboxyl group on oxidation. Dilute nitric acid or chromic acid or alkaline permanganate can be used as oxidant.



Benzoic acid is obtained on oxidation of many compounds like aralkanes (toluene, ethylbenzene, n-propylbenzene, cumene, etc.) and their oxygenated derivatives (benzyl alcohol, benzaldehyde, acetophenone, benzophenone, etc.) halogenated derivatives (benzyl chloride, benzal chloride, benzotrichloride etc) nitrogenated derivatives (cyanobenzene benzamide, etc.)

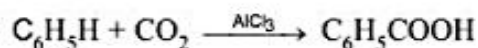
(4) Carboxylation of Grignard's Reagents :

Benzoic acid is formed on hydrolysis of the product obtained by passing carbon dioxide in ethereal solution of phenylmagnesium bromide.



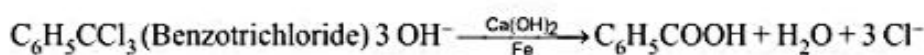
(5) Carboxylation of Benzene :

Benzoic acid is obtained on heating benzene and carbon dioxide in nitrobenzene medium in the presence of anhydrous aluminium chloride.



(6) Alkaline Hydrolysis of Benzotrichloride :

Benzoic acid is formed on hydrolysis of benzotrichloride by lime water in the presence of iron catalyst.

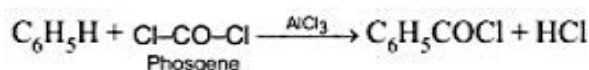


(7) Hydrolysis of Benzoyl Chloride :

Benzoic acid is obtained by hydrolysis of benzoyl chloride



Benzoyl chloride is prepared by Friedel–Crafts chlororformylation of benzene, Benzoic acid is obtained in 60% yield by the method.



(8) Hydrolysis of Cyanobenzene :

Benzoic acid is formed on hydrolysis of cyanobenzene i.e., benzonitrile by dilute hydrochloric acid.



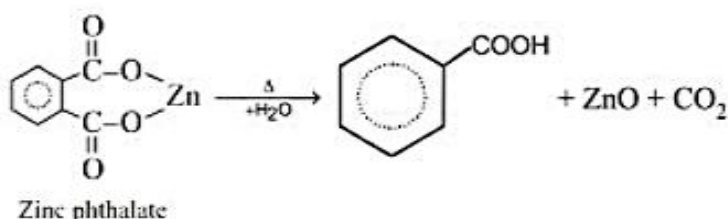
(9) Dehydroxylation of Hydroxybenzoic Acid :

Benzoic acid is formed on dry distillation of o-, m- or p-hydroxybenzoic acid with zinc dust. The ortho isomer is called salicylic acid.



(10) Decarboxylation of Phthalic Acid :

o-Benzendicarboxylic acid is called phthalic acid. On heating its zinc or nickel or chromium salt at 200–300° with water vapours, benzoic acid is formed by partial decarboxylation.



(11) Hydrolysis of Benzoic Acid Derivatives :

Benzoic acid is formed on hydrolysis of esters, anhydride, acid chloride and amide of benzoic acid.



Benzoyl chloride and benzoic anhydride very readily get hydrolysed by water, but with little bit more difficulty as compared to aliphatic acid chlorides. Hydrolysis by alkali takes place faster.



Hydrolysis of benzoic esters takes place on heating with an alkali.



Hydrolysis of benzamide takes place on boiling with dilute acid or alkali solution.



Liberation of ammonia gas on heating the solution after adding sodium hydroxide solution is the test for amides.

(2) Physical Properties

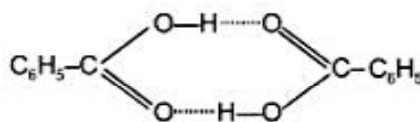
Benzoic acid is a crystalline solid (melting point 122°) which is sparingly soluble in cold water, but readily soluble in hot water. It is also soluble in organic solvents like alcohol ether, etc. Its acid strength is more than that of the alkanolic acids due to weak negative inductive effect of phenyl group.

(3) Resonance in Benzoic Acid

The COOH group present in benzoic acid is a strong electron-attracting group due to which the benzene ring is deactivated and rate of S_E reaction decreases as compared to unsubstituted benzene. Therefore, the COOH group exerts a deactivating effect on the benzene ring.

Dimerisation in Benzoic Acid due to Hydrogen Bonding :

Benzoic acid in liquid phase (in benzene solution or in molten form) exists in the form of a dimer by association due to intermolecular hydrogen bonding in the same fashion as in alkanolic acids. This is the reason why the experimental value of the molecular weight of benzoic acid is twice the calculated value.



Hydrogen bonded dimer of benzoic acid

(4) Reactions

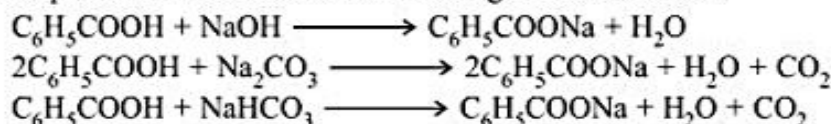
Benzoic acid gives the following two types of reactions.

- (A) Reactions due to COOH group
- (B) Reactions due to C_6H_5 group

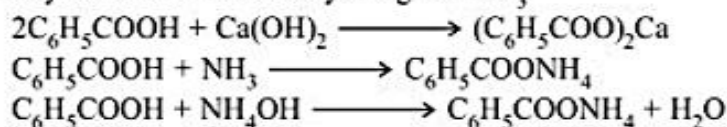
(A) Reaction Due to Carboxyl Group

1. With Bases :

Many alkaline compounds form benzoate salts on reacting with benzoic acid.

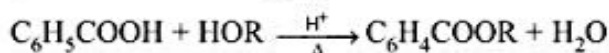


Phenol (a weak acid) does not react with sodium bicarbonate, whereas benzoic acid gives effervescence of CO_2 , although both are acidic organic compounds toward litmus. Therefore, distinction between phenol and carboxylic acids can be done by using NaHCO_3 .



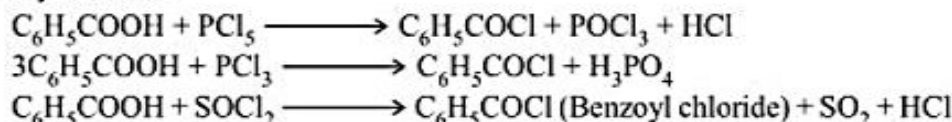
2. With Alcohols :

An alkyl benzoate is formed by esterification on heating a mixture benzoic acid and an alkanol in the presence of a few drops of concentrated H_2SO_4 .



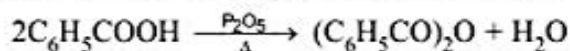
3. With Inorganic Chloride :

Benzoyl chloride is formed on the reaction of benzoic acid with phosphorus pentachloride or phosphorus trichloride or thionyl chloride.



4. With P_2O_5 :

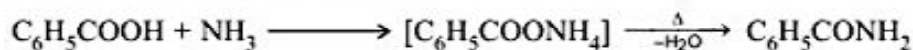
Benzoic anhydride is formed on heating benzoic acid with P_2O_5 .



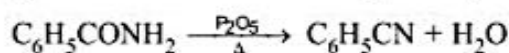
Acetic anhydride also dehydrates two molecules of benzoic acid to give benzoic anhydride.

5. With Ammonia :

Benzamide is formed on heating benzoic acid with ammonia.

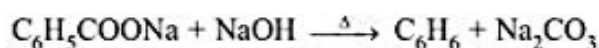


Benzonitrile or cyanobenzene can be obtained by heating benzamide with P_2O_5 .



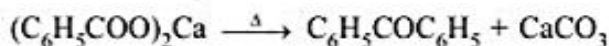
6. With Soda Lime :

Benzene is obtained on dry distillation of sodium salt of benzoic acid with sodalime.



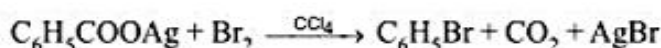
7. Dry Distillation of Calcium Salt :

Benzophenone is obtained on dry distillation of calcium salt of benzoic acid.



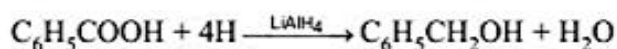
8. Hunsdiecker Reaction :

Bromobenzene is formed on heating the silver salt of benzoic acid with bromine in carbon tetrachloride solution



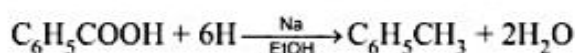
9. Reduction by Lithium Aluminium Hydride :

Benzyl alcohol is formed on reduction of benzoic acid by lithium aluminium hydride (LiAlH_4).



10. Reduction by Sodium and Ethanol :

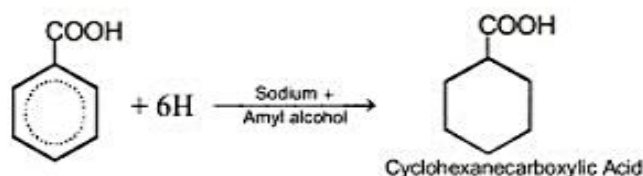
Toluene is formed on reduction of benzoic acid by sodium and ethanol.



(B) Reaction Due to Aromatic Ring

Hydrogenation by Sodium and Amyl Alcohol :

The COOH group is not affected on reduction of benzoic acid by sodium and amyl alcohol but benzene ring of benzoic acid is completely hydrogenated to form Cyclohexanecarboxylic acid.



Aromatic Electrophilic Substitution Reaction :

Meta substitution products are formed on halogenation nitration and sulphonation of benzoic acid.

